

Nicholas W. Fraser

junior **AI engineer** · typescript · cloudflare · llm pipelines

01 / SUMMARY

Junior AI engineer building production TypeScript and LLM applications on Cloudflare.
Founder of **Tek Monkeys**, a Brisbane IT consultancy serving a paediatric orthopaedic practice, where I designed and deployed **Itemate**, a MBS billing validator pairing the Anthropic Claude API with a deterministic matcher and a server-side PHI-free data flow.
Third-year UQ Business (Honours) student, Finance major, graduating 2027.

02 / EXPERIENCE

2024 to present
Brisbane, QLD

Founder and AI Engineer @ Tek Monkeys

brisbane it consultancy · single-client paediatric orthopaedic practice

- Sole technical contact for a paediatric orthopaedic practice; manage IT support, hardware, and software within the practice's obligations under the Privacy Act 1988, the Australian Privacy Principles, and the My Health Records Act 2012.
- Hardware fleet includes a Windows server plus multiple clinical workstations (onsite and remote) and laptops.
- Scoped, designed, and shipped **Itemate**, the practice's MBS billing validator, after workflow interviews with the surgeons. Architecture detailed in [// projects](#).

03 / SELECTED PROJECTS

apr 2026 to present
arcade.nickwfraser.dev

Monkey Barrel **real-time multiplayer web arcade**

TypeScript SvelteKit 5 Cloudflare Workers Durable Objects D1 Web Crypto Bun

- Built and deployed a multiplayer party-game platform hosting 12+ games plus a 6,400-line idle-RPG sub-app (9 zones, 180 stages). Coordinated authoritative game state through Durable Objects on the WebSocket Hibernation API for near-zero idle cost; ~1,000 games played across 14 registered users since launch.
- Architected an actor-model room hierarchy (one Durable Object per room) with PBKDF2 + D1-backed session auth via Web Crypto; diagnosed and hotfixed a production incident where Workers' 100k PBKDF2 iteration cap blocked NIST G1 compliance.
- Caught a Durable Object hibernation regression across 5 room subclasses by running parallel Opus audit agents through Claude Code (23 findings with file-and-line precision in ~10 min); shipped fix as a unified connection model with 18 integration tests passing.

feb 2026
gocard.nickwfraser.dev

GoCard Insights **QLD transit + mode-choice analytics**

TypeScript SvelteKit 5 CF Pages + Workers D1 + KV Chart.js MapLibre GL

- Built and shipped the BSAN4204 Business Analytics research project, combining Go Card trip-history analysis with a live mode-choice calculator comparing public transport against driving costs after Translink's 50¢ flat-fare reform.
- Engineered a daily fuel-data pipeline as a sibling Cloudflare Worker cron against the QLD FPQ API with KV caching and D1 history retention; validated a pure mode-choice engine across 24 unit tests to within \$0.50 per week of manual spreadsheet calculations.

Itemate MBS billing validator

- TypeScript
- SvelteKit 5
- CF Pages + D1
- Drizzle
- Anthropic Claude API
- Vitest

- Built and deployed Itemate for a paediatric orthopaedic practice (two surgeons), pairing the Anthropic Claude API (Haiku for dev, Sonnet for prod) for clinical note normalisation with a deterministic keyword-overlap retrieval pass over a 6,039-item MBS index. Deployed at the edge behind Cloudflare Access (Google OAuth).
- Engineered a regex-anchored de-identification pipeline that strips PHI server-side before any LLM call; shipped 97 unit tests with GitHub Actions CI. No patient data persists on the server.

04 / EDUCATION

feb 2023 to dec 2027
Brisbane, QLD

Bachelor of Advanced Business (Honours), Finance @ University of Queensland
finance major · business analytics electives

- **Relevant coursework:** Machine Learning for Business (BSAN2205), Data Visualisation (BSAN3204), Corporate Finance, Asset Pricing, Derivatives and Risk Management.

05 / TECHNICAL SKILLS

LANGUAGES	TypeScript · JavaScript · Python · SQL · R
AI ENGINEERING	Anthropic Messages API (structured outputs, Haiku/Sonnet routing) · Claude Code · agent orchestration via subagents · prompt engineering · hybrid LLM + deterministic architectures · PHI handling in regulated AI pipelines
FRAMEWORKS	SvelteKit 5 (runes) · Svelte · React 19 · Vite · Node · Bun · Vitest · R/Shiny · tidyverse · pandas · Chart.js · MapLibre GL
PLATFORMS	Cloudflare Workers · Pages · Durable Objects · D1 · KV · Cron Workers · Access · Drizzle ORM · Web Crypto + Web Audio API · Git · GitHub Actions CI
DATA & FINANCE	Excel financial modelling (DCF, sensitivity) · derivatives pricing · interactive dashboards
HEALTHCARE COMPLIANCE	Designed Itemate's PHI-free data flow under the Privacy Act 1988 + Australian Privacy Principles · working knowledge of the My Health Records Act 2012 + MBS billing under the Health Insurance Act 1973

06 / EXTRACURRICULARS & ADDITIONAL

SOCIETIES	UQ Economics Society + Business Analytics Network Association (member since 2023) · attended IB, deal advisory, and analytics speaker series
LANGUAGES	English (native) · Japanese (JLPT N4)
VOLUNTEERING	Community BBQ service for people experiencing homelessness (2020 to 2022) · charity fundraising